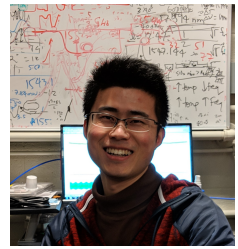


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in Yizhi Luo



Education

- 04/2025 – present **Postdoc, Applied Physics, Stanford University**, United States
Research topic: *Quantum acoustics, superconducting qubit...*
Advisor: *Amir Safavi-Naeini*
- 08/2017 – 03/2025 **Ph.D., Applied Physics, Yale University**, United States
Thesis title: *High-coherence phononic resonator for quantum acoustic applications*
Advisor: *Peter T. Rakich*
- 09/2015 – 06/2017 **M.Sc., Chemistry, University of Tokyo**, Japan
Thesis title: *Rapid-scan Fourier-Transform coherent anti-Stokes Raman scattering spectroscopy*
Advisor: *Keisuke Goda*
- 09/2011 – 06/2015 **B.S., Optical Engineering, Zhejiang University**, China
Thesis title: *Upconversional photoluminescence of CdSe/CdS core-shell colloidal quantum dots*
Advisor: *Xiaogang Peng*

Skills

- Languages
 - Mandarin:** Native
 - English:** Professional
 - Japanese:** Basic
 - French:** Basic
- Optical engineering
 - 10 years** of working experience in optical labs, developing nonlinear laser spectroscopy, building free-space and fiber optical setups for sensing and precision measurements.
- Coding
 - 5 years** of **Python** programming for optical/acoustic wave simulation and experimental data analysis.
- Cryogenic engineering
 - 3 years** of using and debugging various cryogenic system such as Bluefors dilution refrigerator and Janis helium recirculating gas cooling system.
- Cleanroom fab
 - 6 years** of cleanroom experience of fabricating on-chip acoustic resonator devices.
- Material analysis
 - grazing-incidence XRD, AFM, optical profilometer, etc.
- Hobbies
 - badminton, tennis (single-handed backhand), poetry and language studies.

Selected Awards

- 2017 **Yale Graduate Fellowship**, Yale University
- 2016 **Advanced Leading Graduate Course for Photon Science (ALPS) program**, University of Tokyo

Selected Awards (continued)

- 2015  **Excellent Student Thesis Awards**, Zhejiang University
- 2014  **Excellent Student Awards**, Zhejiang University
-  **First Prize of College Students' Physics Innovation Competition**, Zhejiang Province

Research experience

- 04/2025 – present  **Postdoc**, Stanford University
Research topic: quantum acoustics, superconducting qubit
Study phonon decoherence mechanisms in micro-fabricated phononic resonators and their interaction with superconducting qubit at cryogenic temperature.
- 08/2017 – 03/2025  **PhD program**, Yale University
Research topic: optomechanics, nonlinear optics
- **Micro-fabrication of high-overtone bulk acoustic resonators**
During my PhD years, I've been working on building record-high-quality phononic resonators for quantum applications. We developed a reflow-based technique that enables scalable fabrication of high-overtone bulk acoustic resonators (HBARs) on high-purity crystalline quartz substrates.
 - **Build a numerical eigenmode solver to calculate the acoustic eigenmodes for fabricated HBARs**
During the pandemic lockdown (2020-2021), I developed an eigenmode solver for the HBARs. The simulation algorithm consists of Fourier beam propagation method and eigenmode analysis. It can calculate the mode profiles, eigen-frequencies and geometric losses of millimeter-scale resonators that are usually impossible to simulate using the traditional FEM or FDTD methods.
 - **Coherent anti-Stokes Brillouin scattering (CABS) spectroscopy**
I built a coherent anti-Stokes Brillouin scattering (CABS) setup as a non-invasive laser-based tool to study phonon coherence, featuring the optomechanical interaction. This technique enables measuring both the phonon lifetime and the coherence time.
 - **Material characterization of HBAR substrates**
A few powerful material analysis methods have been used to figure out the limiting factors of phonon losses in our HBAR devices, which involves careful measurements of bulk defects and surface/sub-surface imperfection. These include the optical profilometer, atomic force microscope and grazing-incidence X-ray rocking-curve measurement, and secondary ion mass spectroscopy for impurity concentration.
- 09/2015 – 06/2017  **Master program**, University of Tokyo
Research topic: ultrafast optics, nonlinear optics, Raman scattering
During my master program in University of Tokyo, I built a broadband Fourier-transform coherent anti-Stokes Raman spectrometer using a Ti:Sapphire femtosecond laser, with enhanced signal-to-noise ratio from heterodyne detection. By integrating a fast scanning mirror (12 kHz), it can realize spectral acquisition rate of >20,000 scans/sec. It is then implemented in fast biological cell detection and sorting.

Publications

Journal Articles

- 1 H. H. Diamandi, **Y. Luo**, D. Mason, *et al.*, “Optomechanical control of long-lived bulk acoustic phonons in the quantum regime,” *Nature Physics*, vol. 21, no. 9, pp. 1482–1488, 2025.
- 2 **Y. Luo**, H. H. Diamandi, H. Li, *et al.*, “Lifetime-limited gigahertz-frequency mechanical oscillators with millisecond coherence times,” *arXiv preprint arXiv:2504.07523*, 2025.
- 3 I. Kudelin, W. Groman, Q.-X. Ji, *et al.*, “Photonic chip-based low-noise microwave oscillator,” *Nature*, vol. 627, no. 8004, pp. 534–539, 2024.
- 4 N. Jin, C. A. McLemore, D. Mason, *et al.*, “Micro-fabricated mirrors with finesse exceeding one million,” *Optica*, vol. 9, no. 9, pp. 965–970, 2022.
- 5 C. A. McLemore, N. Jin, M. L. Kelleher, *et al.*, “Miniaturizing ultrastable electromagnetic oscillators: Sub-10-14 frequency instability from a centimeter-scale fabry-pérot cavity,” *Physical Review Applied*, vol. 18, no. 5, p. 054054, 2022.
- 6 K. Hiramatsu, T. Ideguchi, Y. Yonamine, *et al.*, “High-throughput label-free molecular fingerprinting flow cytometry,” *Science Advances*, vol. 5, no. 1, eaau0241, 2019.
- 7 K. Hiramatsu, **Y. Luo**, T. Ideguchi, and K. Goda, “Rapid-scan fourier-transform coherent anti-stokes raman scattering spectroscopy with heterodyne detection,” *Optics Letters*, vol. 42, no. 21, pp. 4335–4338, 2017.

Conference Papers

- 1 **Y. Luo**, T. Yoon, D. Mason, *et al.*, “Studies of high-q phononic resonator for quantum acoustic applications,” in *APS March Meeting Abstracts*, vol. 2023, 2023, Q66–006.
- 2 **Y. Luo**, D. Mason, J. P. Hendrie, *et al.*, “Numerically accelerated development cycle for ultra-high finesse micro-fabricated resonators,” in *CLEO: Science and Innovations*, Optica Publishing Group, 2021, JW1A–17.